20. Wednesday, May 14. Cartesian Products (5.4), Indexed families of sets (5.5)

Last Time: For sets A, B defined ordered pairs (a, b)

Definition 1 Let A and B be sets. The **Cartesian product** $A \times B$ is the set of all possible ordered pairs (a, b) where $a \in A$ and $b \in B$. That is:

$$A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$$

"A cross B" — A times B

Notation: $A^2 = A \times A$.

Ex: $A = \{1,2,3\}$, $B = \{a, b\}$. Then

$$A \times B = \{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b)\} \quad \text{Size } 6 = 3 \cdot 2$$

Note $(a,1) \notin A \times B$, but $(a,1) \in B \times A$.

Theorem 2 Let A, B, C sets. Then $(A \cup B) \times C = (A \times C) \cup (B \times C)$

Proof We will prove both inclusions by element-chasing.

[$\subseteq$] Suppose $(x,y) \in (A \cup B) \times C$. Then $x \in A \cup B$ and $y \in C$.

Since $x \in A \cup B$, we have $x \in A$ or $x \in B$. There are two cases.

Case 1: If $x \in A$, then since $y \in C$, we have $(x,y) \in A \times C$, hence $(x,y) \overset{\text{thm 19.1}}{\in} A \times C \cup B \times C$

Case 2: If $x \notin A$, since $x \in A \cup B$, we have $x \in B$. Since $y \in C$, we have $(x,y) \in B \times C$

Then also $(x,y) \in (A \times B) \cup (A \times C)$

This shows $(A \cup B) \times C \subseteq (A \times C) \cup (B \times C)$

[$\supseteq$] Suppose $(x,y) \in (A \times C) \cup (B \times C)$. Then $(x,y) \in A \times C$ or $(x,y) \in B \times C$.

Case 1: If $(x,y) \in A \times C$, by definition, $x \in A$ and $y \in C$. Since $x \in A \subseteq A \cup B$, we have $(x,y) \in (A \cup B) \times C$

Case 2: If $(x,y) \notin A \times C$, Then $(x,y) \in B \times C$. Thus $x \in B$ and $y \in C$. Hence $x \in A \cup B$, it follows $(x,y) \in (A \cup B) \times C$.

This shows $(A \times C) \cup (B \times C) \subseteq (A \cup B) \times C$

We have shown both inclusions, so the sets are equal $\square$

Indexed families of sets (5.5)

Last Time (Part of AST)

Given sets A, B, C, we know $(A \cup B) \cup C = A \cup (B \cup C)$

This means we can just define the union of these sets to be

$$A \cup B \cup C \overset{\text{def}}{=} (A \cup B) \cup C = A \cup (B \cup C)$$

And can similarly define the union of any finite family of sets

Even better, can do the same for any collection of sets.

Definition 3: Let $\mathcal{C}$ be a family of sets (could be infinite)

- The **union over $\mathcal{C}$** is the set of all elements that are in at least one of the sets in $\mathcal{C}$

Notation: $\bigcup_{A \in \mathcal{C}} A = \{x \mid x \in A \text{ for some } A \text{ in } \mathcal{C}\}$.

\bigcup_{A \in \mathcal{C}}

- The **intersection over $\mathcal{C}$** is the set of all elements that are in all of the sets in $\mathcal{C}$

Notation: $\bigcap_{A \in \mathcal{C}} A = \{x \mid x \in A \text{ for some } A \text{ in } \mathcal{C}\}$.

\bigcap_{A \in \mathcal{C}}

Q: How do we work with such $\mathcal{C}$?

Frequently, there is an index set Λ such that we have a set A_α in $\mathcal{C}$ for each $\alpha \in \Lambda$

We write this as $\mathcal{C} = \{A_\alpha \mid \alpha \in \Lambda\}$ and $\bigcup_{A \in \mathcal{C}} A = \bigcup_{\alpha \in \Lambda} A_\alpha$, same for $\bigcap$.

For example, we could have

$\mathcal{C} = \{A_1, A_2, A_3, A_4, \ldots\} = \{A_i \mid i \in \mathbb{N}\}$ so $\Lambda = \mathbb{N}$ here

Indexed by $\mathbb{N}$

In this case we write $\bigcup_{A \in \mathcal{C}} A = \bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i=1}^{\infty} A_i$

$\bigcup_{i=1}^{\infty}$ → Only for $\Lambda = \mathbb{N}$

Example: For each $n \in \mathbb{N}$, let $I_n = [0, n) \subseteq \mathbb{R}$, and let $\mathcal{C} = \{I_1, I_2, I_3, I_4\}$

So index set is $\{1,2,3,4\}$

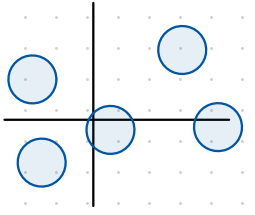
Then $\bigcap_{I \in \mathcal{C}} I = [0, 1) = I_1$,

$\bigcup_{I \in \mathcal{C}} I = [0, 4) = I_4$

But if we change our $\mathcal{C}$, and take $\mathcal{D} = \{I_n \mid n \in \mathbb{N}\}$ so index set is $\mathbb{N}$

Then $\bigcup_{I \in \mathcal{D}} I = \bigcup_{n \in \mathbb{N}} I_n = [0, \infty) = \{x \in \mathbb{R} \mid x \geq 0\}$

Example: In $\mathbb{R}^2$, let $\mathcal{C} = \{$ Unit disks (i.e. radius 1) with any center $\}$



How can we index $\mathcal{C}$?

For each $(a,b) \in \mathbb{R}^2$ let

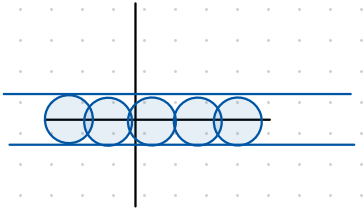
$D(a,b)$ = Unit disk with center (a,b)

So $\mathcal{C} = \{ D(a,b) \mid (a,b) \in \mathbb{R}^2 \}$.

Of course $\bigcup_{(a,b) \in \mathbb{R}^2} D(a,b) = \mathbb{R}^2$

More interesting:

$$\bigcup_{a \in \mathbb{R}} D(a,0) = \text{Horizontal strip between } -1 \text{ and } 1$$



Theorem 4: Let Λ be a non-empty index set. Let $\{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets. Then:

1) For each $\beta \in \Lambda$, $A_\beta \subseteq \bigcup_{\alpha \in \Lambda} A_\alpha$

2) For each $\beta \in \Lambda$, $\bigcap_{\alpha \in A} A_\alpha \subseteq A_\beta$

3) DeMorgan's laws: $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$ and $\left(\bigcap_{\alpha \in A} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$

Proof: Let's proof the second DeMorgan's law. We prove both inclusions.

[$\subseteq$] Let $x \in \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c$. Then $x \notin \bigcap_{\alpha \in \Lambda} A_\alpha$. Since x is <u>not</u> in the intersection, by def^n this means $\exists\, \beta \in \Lambda$ such that $x \notin A_\beta$. Thus $x \in A_\beta^c$, Therefore $x \in \bigcup_{\alpha \in \Lambda} A_\alpha^c$ by definition.

[$\supseteq$] Suppose $x \in \bigcup_{\alpha \in \Lambda} A_\alpha^c$. By def^n, $\exists\, \beta \in \Lambda$ such that $x \in A_\beta^c$. Hence $x \notin A_\beta$.

By definition of intersection, $x \notin \bigcap_{\alpha \in \Lambda} A_\alpha$. Hence $x \in \left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c$

The other statements are exercises $\square$